

Contract CO1: createTask

Operation: createTask(title: String, desc: String)

Cross References: Use Case: Create Task

Preconditions:

- title is not empty.

Postconditions:

- A Task instance t was created. (instance creation)
- t.title was set to title. (attribute modification)
- t.description was set to desc or empty. (attribute modification)
- t.creationDate was set to the current timestamp. (attribute modification)
- t.status was set to "open". (attribute modification)
- t was associated with TaskDB. (association formed)
- An ActivityEntry instance ae was created. (instance creation)
- ae.timestamp was set to current timestamp. (attribute modification)
- ae.description was set to "Task created". (attribute modification)
- ae was associated with t. (association formed)
- ae was associated with ActivityHistory. (association formed)

Contract CO2: addDescription

Operation: addDescription(desc: String)

Cross References: Use Case: Update Task

Preconditions:

- A Task instance t exists.

Postconditions:

- t.description was set to desc. (attribute modification)
- An ActivityEntry instance ae was created. (instance creation)
- ae.description was set to "Task description updated". (attribute modification)
- ae was associated with t. (association formed)

Contract CO3: setPriority

Operation: setPriority(priority: PriorityLevel)

Cross References: Use Case: Update Task

Preconditions:

- A Task instance t exists.

Postconditions:

- t.priority was set to priority. (attribute modification)
- An ActivityEntry instance ae was created. (instance creation)
- ae was associated with t. (association formed)

Contract CO4: setDueDate

Operation: setDueDate(date: Date)

Cross References: Use Case: Update Task

Preconditions:

- A Task instance t exists.

Postconditions:

- t.dueDate was set to date. (attribute modification)
- An ActivityEntry instance ae was created. (instance creation)
- ae was associated with t. (association formed)

Contract CO5: updateTaskTitle

Operation: updateTaskTitle(t: Task, title: String)

Cross References: Use Case: Update Task

Preconditions:

- Task instance t exists.
- title is not empty.

Postconditions:

- t.title was set to title. (attribute modification)
- An ActivityEntry instance ae was created. (instance creation)
- ae was associated with t. (association formed)

Contract CO6: completeTask

Operation: completeTask(t: Task)

Cross References: Use Case: Complete Task

Preconditions:

- Task instance t exists.
- t.status is not "completed".

Postconditions:

- t.status was set to "completed". (attribute modification)
- An ActivityEntry instance ae was created. (instance creation)
- ae.description was set to "Task completed". (attribute modification)
- ae.timestamp was set to current timestamp. (attribute modification)
- ae was associated with t. (association formed)

Contract CO7: cancelTask

Operation: cancelTask(t: Task)

Cross References: Use Case: Cancel Task

Preconditions:

- Task instance t exists.

Postconditions:

- t.status was set to "cancelled". (attribute modification)
- An ActivityEntry instance ae was created. (instance creation)
- ae.description was set to "Task cancelled". (attribute modification)
- ae was associated with t. (association formed)

Contract CO8: assignTaskToProject

Operation: assignTaskToProject(t: Task, p: Project)

Cross References: Use Case: Update Task

Preconditions:

- Task instance t exists.
- Project instance p exists.

Postconditions:

- t was associated with p. (association formed)

Contract CO9: removeTaskFromProject

Operation: removeTaskFromProject(t: Task)

Cross References: Use Case: Update Task

Preconditions:

- Task instance t exists.
- t is associated with a Project.

Postconditions:

- Association between t and its Project was removed. (association removed)

Contract CO10: viewTasks

Operation: viewTasks()

Cross References: Use Case: View Tasks

Preconditions:

- None.

Postconditions:

- Task instances were retrieved from TaskDB. (association traversal)
- No instances were created, modified, or deleted.

Contract CO11: viewTasksByPriority

Operation: viewTasksByPriority(priority: PriorityLevel)

Cross References: Use Case: View Tasks

Preconditions:

- None.

Postconditions:

- Task instances matching priority were retrieved. (association traversal)

Contract CO12: viewTasksByStatus

Operation: viewTasksByStatus(status: Status)

Cross References: Use Case: View Tasks

Preconditions:

- None.

Postconditions:

- Task instances matching status were retrieved. (association traversal)

Contract CO13: viewTasksByProject

Operation: viewTasksByProject(p: Project)

Cross References: Use Case: View Tasks

Preconditions:

- Project instance p exists.

Postconditions:

- Tasks associated with p were retrieved. (association traversal)

Contract CO14: viewTasksByTag

Operation: viewTasksByTag(tag: Tag)

Cross References: Use Case: View Tasks

Preconditions:

- Tag instance tag exists.

Postconditions:

- Tasks associated with tag were retrieved. (association traversal)

Contract CO15: viewTasksByDateRange

Operation: viewTasksByDateRange(startDate: Date, endDate: Date)

Cross References: Use Case: View Tasks

Preconditions:

- startDate and endDate are valid.

Postconditions:

- Tasks within date range were retrieved. (association traversal)

Contract CO16: searchTasksByKeyword

Operation: searchTasksByKeyword(keyword: String)

Cross References: Use Case: Search by Keyword

Preconditions:

- keyword is not empty.

Postconditions:

- Task instances matching keyword in title or description were retrieved. (association traversal)

Contract CO17: createSubtask

Operation: createSubtask(parent: Task, title: String)

Cross References: Use Case: Create Subtask

Preconditions:

- Task instance parent exists.
- title is not empty.

Postconditions:

- A Subtask instance s was created. (instance creation)
- s.title was set to title. (attribute modification)
- s.completionStatus was set to "open". (attribute modification)
- s was associated with parent. (association formed)

Contract CO18: completeSubtask

Operation: completeSubtask(s: Subtask)

Cross References: Use Case: Complete Subtask

Preconditions:

- Subtask instance s exists.

Postconditions:

- s.completionStatus was set to "completed". (attribute modification)

Contract CO19: cancelSubtask

Operation: cancelSubtask(s: Subtask)

Cross References: Use Case: Cancel Subtask

Preconditions:

- Subtask instance s exists.

Postconditions:

- s.completionStatus was set to "cancelled". (attribute modification)

Contract CO20: updateSubtaskTitle

Operation: updateSubtaskTitle(s: Subtask, title: String)

Cross References: Use Case: Update Subtask

Preconditions:

- Subtask instance s exists.

Postconditions:

- s.title was set to title. (attribute modification)

Contract CO21: createProject

Operation: createProject(title: String, desc: String)

Cross References: Use Case: Create Project

Preconditions:

- title is not empty.

Postconditions:

- A Project instance p was created. (instance creation)
- p.name was set to title. (attribute modification)
- p.description was set to desc. (attribute modification)
- p was associated with ProjectDB. (association formed)

Contract CO22: updateProject

Operation: updateProject(p: Project, title: String, desc: String)

Cross References: Use Case: Update Project

Preconditions:

- Project instance p exists.

Postconditions:

- p.name was set to title. (attribute modification)
- p.description was set to desc. (attribute modification)

Contract CO23: completeProject

Operation: completeProject(p: Project)

Cross References: Use Case: Complete Project

Preconditions:

- Project instance p exists.

Postconditions:

- Project completion status was set to "completed". (attribute modification)

Contract CO24: cancelProject

Operation: cancelProject(p: Project)

Cross References: Use Case: Cancel Project

Preconditions:

- Project instance p exists.

Postconditions:

- Project completion status was set to "cancelled". (attribute modification)

Contract CO25: createTag

Operation: createTag(name: String)

Cross References: Use Case: Create Tag

Preconditions:

- name is not empty.

Postconditions:

- A Tag instance tag was created. (instance creation)
- tag.name was set to name. (attribute modification)
- tag was associated with TagDB. (association formed)

Contract CO26: updateTag

Operation: updateTag(tag: Tag, name: String)

Cross References: Use Case: Update Tag

Preconditions:

- Tag instance tag exists.

Postconditions:

- tag.name was set to name. (attribute modification)

Contract CO27: deleteTag

Operation: deleteTag(tag: Tag)

Cross References: Use Case: Delete Tag

Preconditions:

- Tag instance tag exists.

Postconditions:

- tag was removed from TagDB. (association removed)
- tag was disassociated from all Task instances. (association removed)

Contract CO28: assignTagToTask

Operation: assignTagToTask(t: Task, tag: Tag)

Cross References: Use Case: Update Task

Preconditions:

- Task instance t exists.
- Tag instance tag exists.

Postconditions:

- tag was associated with t. (association formed)

Contract CO29: removeTagFromTask

Operation: removeTagFromTask(t: Task, tag: Tag)

Cross References: Use Case: Update Task

Preconditions:

- association exists between t and tag.

Postconditions:

- association between t and tag was removed. (association removed)

Contract CO30: viewTaskHistory

Operation: viewTaskHistory(t: Task)

Cross References: Use Case: View Task Activity History

Preconditions:

- Task instance t exists.

Postconditions:

- ActivityEntry instances associated with t were retrieved. (association traversal)
- No instances were created, modified, or deleted.